

munication Union (ITU), and the International Standards Organization (ISO)—are developing standards for QKD. This will eventually allow multi-vendor interoperability and open new development opportunities. ITU has come up with a series of standards for the QKD network. Also, a few standards are under preparation.

The ITU-T standards in force are:

- Y.3800: Overview on networks supporting quantum key distribution
- Y.3801: Functional requirements for quantum key distribution networks
- Y.3802: Quantum key distribution networks—Functional architecture
- Y.3803: Quantum key distribution networks—Key management
- Y.3804: Quantum key distribution networks—Control and management

Application

There has been a substantial increase in the internet population along with the number of devices per capita and average data usage. With the roll-out of 5G networks, a much greater number of devices are expected to be connected to the internet. Data security will be more challenging and crucial than ever. As per the report from The Insight Partners titled 'Quantum Cryptography Market To 2027,' the Global QKD market is expected to touch \$2 billion by 2027 with a CAGR of 39.2%.

The induction of QKD systems in existing networks to make them quantum secure does not require the entire network to be redesigned. The existing infrastructure can be upgraded with quantum security. The government of India in its Budget 2020 announced a National Mission on Quantum Technologies & Applications (NM-QTA) with a total budget outlay of ₹8,000 crore (\$1.2 billion approx.) for five years to be implemented by the Department of Science & Technology (DST).

QKD systems have their rel-

evance in all the networks that carry confidential, sensitive, financially important, or strategic data. Leakage of this data to a malicious user can cause financial, reputational, or strategic loss. The potential users are:

- Banking, financial services, and insurance (BFSI) sectors
- Networks encompassing government offices located across the country
- Healthcare and pharmaceutical industries
- Military and defence networks
- Telecommunications and the IT industry

Contribution of C-DOT in QKD

C-DOT is a premier telecom R&D institute under the Department of Telecommunications, government of India. C-DOT has the legacy of delivering cutting-edge indigenous technology, hardware, and software solutions that are especially targeted to meet the requirements of the rural areas that were digitally unserved for a long time. In the field of QKD, C-DOT has achieved the following milestones:

- Indigenously developed complete hardware and software for both Alice and Bob nodes using both DPS and COW QKD protocols
- Achieved a secure key rate of greater than 1kbps over 100km fibre with a quantum bit error rate (QBER) of 5%
- Successfully simulated photon number splitting (PNS) attack
- Filed two patents with Indian Patent Office:

(i) Device and Method of Synchronisation in Quantum Key Distribution (QKD) System

(ii) An Efficient Method to Automatically Align Phase Modulator with Pulse Generator/Intensity Modulator in the Alice Node of DPS-QKD System

- Integrated the QKD system with C-DOT router as well as post-quantum cryptography (PQC) system as per ETSI standard **EFY**



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